

Guidelines for Replicators - Secondary Data Replications

This is a living document that may be updated periodically.

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Overview

The credibility of scientific findings is essential to public trust and evidence-based decision-making. In health research, published studies shape policy, guide clinical practice, and influence funding priorities, underscoring the importance of credible findings in those literatures. Although replication studies can be resource-intensive, they play a critical role in strengthening the evidence base. By re-examining the same research questions using new data, replication efforts can contribute to theoretical understanding, reduce uncertainty, and increase confidence in reported conclusions.

The Center for Open Science (COS) is launching the **Replicability Project: Health Behavior (RPHB)**, a large-scale, multi-team effort to replicate a diverse sample of published quantitative studies to help support a transparent and trustworthy foundation in health research. The project aims to assess the robustness of findings and promote best practices in research transparency, methodological rigor, and reproducibility. We welcome individual scholars, labs, and research teams across all career stages and institutions to join the effort. This project is supported by grant funding from the Robert Wood Johnson Foundation and XTX Markets.

What studies are being replicated?

Replications will focus on a sample of studies published between 2015 and 2024 in the following journals:

- Journal of Health Communication (ISSN: 1081-0730)
- Social Science & Medicine (ISSN: 0277-9536)
- Journal of Public Health (ISSN: 1741-3842)
- Journal of Public Health: From Theory to Practice (ISSN: 0943-1853)
- Applied Research in Quality of Life (ISSN: 1871-2576)
- American Journal of Health Promotion (ISSN: 0890-1171)
- Annals of Behavioral Medicine (ISSN: 0883-6612)

Key Project Details

A secondary data replication should provide a good faith, high-quality test of the focal claim with data that was ***not used*** in the original study. As with 'new data' replications, recreating the exact conditions of the original data and analyses is not possible, and that should not be the priority. Rather, the priority is identifying data sources and analyses that provide a sound test of the underlying theoretical claim even with potential differences in samples, measures, and data structures. The operative question that drives the design and evaluation of all replications, including secondary data cases, is whether the replication study will result in diagnostic

evidence about the original claim – that is, would confirming evidence strengthen our confidence in the original claim, and would disconfirming evidence weaken our confidence in it.

Secondary data replications present unique constraints that are worth highlighting. Replication analysts and reviewers should be sure to attend to these issues in the preregistration and final reporting.

- Population: Whether and 'by how much' the replication population can differ from the original study's population should be determined by theoretical framing in the original article. When authors tie their theory or findings to a specific country, region, or subpopulation, replication studies would need to be conducted on the same population to be considered good faith. Conversely, theoretical claims that are implicitly or explicitly framed as more general can allow a wider range of populations for re-testing.
- Measures: Secondary data analysts have less control over the availability and operationalization of key variables compared to new data replicators. This need not be disqualifying for a potential replication study, but replicators should comment explicitly on the construct validity of the main predictor and outcome variables in the replication compared to the original study. Additionally, if covariates from the original study are not available in the replication data, replicators should justify why the replication analysis is still a good faith and high quality test in the absence of these controls.
- Data overlap: A key challenge of secondary data replications is potential non-independence in the units being studied. This arises most often in the case of panel data studies, where later waves of an existing panel dataset will often include many of the same individuals whose data was used in the original study. For a panel dataset to be a valid replication dataset, it needs to be sufficiently powered for the replication analysis ***after excluding*** any individuals whose earlier wave data was already used in the original article.

Contact info

Please direct all questions and project updates to Ella Boycan or Ashleigh Blersch at replications@cos.io. Ella and Ashleigh will also be in touch on a regular basis to help track your progress.

Project deadline

All replication studies must be completed and reported to COS by **December 1, 2026**.

Funding

The Center for Open Science is offering **$750 USD** for each complete secondary data replication study. We are currently limiting each research team to a maximum of three replication studies.

Transparent research practices

Replicators will be asked to preregister their replications, upload all of their materials and outputs to a repository of their choice, link their outputs to their preregistration, and report their outcomes to the project team. COS employees will be available for guidance with carrying out these open science practices.

Replication definition

We use the term *replication*, in the context of this project, to mean investigating the same empirical claim from an original study in a new sample (i.e., by collecting new data, or by using secondary data sources independent from the data used in the original study). More broadly, replication can be understood as a study for which any outcome would be considered diagnostic evidence about a claim from prior research. The purpose of replication is to advance theory by confronting existing understanding with new evidence. See this article for a more comprehensive discussion of replication.

Process and Timeline

Power Estimation

One of the very first steps for the replication team will be to conduct an *a priori* power analysis in order to calculate the approximate power given the sample size of the existing dataset(s) used for the replication. Please review the Secondary Data Power Analysis Guidance document, which provides general guidance as well as the target criteria for the power analysis. Return your power analysis documentation and target sample size to replications@cos.io **ASAP** (and let us know if you have any questions/need help with this).

Preregistration

Begin planning your study design, methodology, and analysis plan by completing a preregistration. Please review the Preregistration Instructions for Replicators and the sample template to get a sense of what kind of information you'll be asked to provide, though note that this is just a sample version so don't begin to fill it out just yet! We will send you your own semi-prefilled document to complete.*

*Please **do not** make a separate copy of the prereg doc that we send you, or download a word doc copy to work within separately. Please fill out and continue to work within the same google doc that we provide to you, as we would like to maintain this doc as the version of record (to help minimize the potential for several versions floating about, or someone inadvertently losing access to the doc). But feel free to use a separate doc to draft content if you would like, just be sure to transfer any content or updates over to the google doc prereg. We will add you and any others directly to the google doc so you have full access, but if you do not have a google account please let us know and we can provide edit access via a sharable-link.

- All preregistration drafts should be completed no later than **two weeks** after we provide you with your preregistration document.

Testing your analysis code: The preregistration for an existing data replication should include all of the data processing and analysis code that you will use to complete your study. You should test your analysis code on a random 5% of the replication sample to confirm it works as intended, and you should include evidence that the code is functional in your preregistration and on the OSF project. Please do not run the analysis code on the full sample until the preregistration has received editorial approval and you have registered your project on the OSF. This ensures the preregistration and review process remain results-blind.

Preregistration Review

Once you have completed your preregistration document, your preregistration will be reviewed by an external team – each preregistration will be reviewed by one Editor and one to two

Reviewers. The aim of this review workflow is to maximize the quality of the replication studies and the clarity and completeness of the preregistrations through a **rapid collaborative review process**. For the purposes of this replication project, a preregistration is considered high quality if it generates a protocol that is a good faith attempt to replicate the original finding.

Review Timeline: The preregistration review period should last approximately **7 days**.

1) Once the replication team completes their preregistration draft, COS alerts the review pool that another preregistration is ready for review and adds it to the queue of preregs waiting for review. We will aim to initiate review **within 1-2 weeks** of the prereg draft being complete (however the time it takes to initiate review depends on how many preregs are waiting for review and reviewer pool capacity at the time).
2) Once the review is initiated, the review period stays open for **7 days**. The replication team addresses comments/suggestions from reviewers (+ any original authors) and makes necessary changes within the Google Doc **in real time** during the review. The Editor oversees the review, weighing in on unresolved comments and assisting to resolve any issues that crop up as necessary.
3) The review period closes after **7 days.** The process concludes once the Editor signs off on the final version of the preregistration.

Please see the RPHB Preregistration Review Guide for additional details.

OSF Registration

- Once the researcher has received editorial approval of their preregistration they will need to preregister their project on the OSF by following the steps linked below. After selecting the replication's OSF project then selecting the Secondary Data Preregistration form, the researcher will simply copy their responses from the finalized preregistration document into the form on the OSF, then submit their registration.
    - Help with Preregistrations
- Researchers should aim to preregister their project on OSF within **1 week** and send the preregistration link to replications@cos.io.

Data Preparation & Analysis

- Analysts can begin working on their final analyses once they have registered their study. Please aim to have your final report complete **within three weeks** of registering.
- Final reports must be received by COS by **December 1, 2026** at the very latest.
    - **This is a firm deadline and no extensions will be granted.**

Output Reporting

- At the end of the project, the researcher connects outputs to the preregistration for data, code, and materials.

- Final reports should be filled out and all materials uploaded to OSF, **within 2 weeks** of completing analyses.
- COS will aim to review and get back to the replication team with confirmation or questions/requests **within 1-2 weeks** of your submission (depending on project stage/capacity).
- **Cutoff date:** All final reports must be complete and all materials (including final report) must be uploaded to the OSF project **no later than December 1, 2026.**

FAQs

Claims

Can I change the identified claim from the paper?

- ➔ No, the claims were identified following an established process, and maintaining the same claim is important for the goals of this project.

Sharing Outcomes & Authorship Opportunities

Can we publish our replication outcomes as a preprint? Or submit our study for publication in a journal?

- ➔ Replication teams are encouraged to publish their replication outcomes publicly as a preprint. Teams are also free to submit their replication study for publication in a journal.

Will there be authorship opportunities for this replication effort at large?

- ➔ COS plans to lead a collaborative paper/report to summarize all of the replication outcomes included in this effort. We will invite all researchers who participate in the replication effort to be co-authors on the paper (this includes all replicators, as well as preregistration reviewers and editors).

Statistical Power

How do I estimate the approximate power of my replication study?

- ➔ One of the very first steps for the replication team will be to conduct an *a priori* power analysis in order to calculate the approximate power given the sample size of the existing dataset(s) used for the replication. Please review the Sample Size Calculations for RPHB document, which provides general guidance as well as the target criteria for the power analysis. Return your power analysis documentation and target sample size to replications@cos.io **ASAP**, otherwise please let us know if you have any questions or need help with this.

Resources

- ★ [TEMPLATE V.1 2025] ACKNOWLEDGEMENT OF HONORARIUM
- ★ Secondary data power
- ★ Preregistration Instructions for Replicators
- ★ RPHB Existing Data Preregistration Template
- ★ Instructions for Preregistration Reviewers
- ★ RPHB Prereg Review Criteria for Existing Data Replications
- ★ RPHB Preregistration Review Guide
- ★ Secondary data sourcing
- ★ RPHB Webpage
- ★ What is replication?
- ★ What is preregistration?